



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION
JUNE 23

Course: EC3103 (Introduction Information Systems)

Time: 8.00 am – 11.00 am
(3 hours)

Lecturer: Thagirarani Muniandy

Date: 29 September 2023

Instructions:

Answer ALL questions in the Answer Booklet provided.

The maximum number of marks is 100.

This exam paper consists of 3 printed pages.
(excluding the front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. Differentiate between open, closed, and adaptive systems. Give an example for each type of system. (6 MARKS)
 - b. Describe what is a system with an example. (5 MARKS)
 - c. Briefly explain THREE (3) basic functions of a system. (6 MARKS)
2.
 - a. Briefly discuss the most suitable system conversion technique to be used in each of the following three case studies.
 - i. Mr. Adam, the owner of XY Pictures, is in the process of developing a system that will control the money flow in this production. (3 MARKS)
 - ii. SN Security Private Limited, plans to have a system, called SN System that will be used at all branches in Negeri Sembilan. (3 MARKS)
 - iii. Crackers Private Limited is planning to have a system for her crackers-selling, which integrate all subsystems such as crackers stock, account payable, and account receivable in her company. (3 MARKS)
 - b. Why has prototyping become a popular way to develop business applications? (6 MARKS)
 - c. List the steps in the prototyping process. (5 MARKS)

- 3 a. What is the purpose of knowledge management? (3 MARKS)
- b. Discuss TWO (2) benefits for an organization by using the knowledge management system. (4 MARKS)
- c. Explain THREE (3) competitive strategies enabled by information technology that firms can pursue. Give an example for each competitive strategy. (6 MARKS)

- 4 Tesla Pharmacy, a well-established pharmacy serving the residents of Nepal, plays a vital role in providing a wide range of medicines and healthcare products to the community. Alongside over-the-counter (OTC) medications, Tesla Pharmacy also dispenses controlled drugs, in accordance with the regulations set forth by the Ministry of Health. These regulations mandate that a pharmacist maintains a detailed record, following a standardized format, of all controlled drug transactions, including specific buyer information and particulars of the controlled drugs sold. In response to these requirements, the owner of Tesla Pharmacy has approached you, an IT consultant, to develop a database system capable of meeting these obligations.

The database structure for this project involves several entities. Medications are categorized as either controlled or uncontrolled drugs. Both drug categories share common attributes, including drug ID, drug name, dosage, price, and type. Uncontrolled or OTC drugs have an additional attribute called "purpose," while controlled drugs require details such as prescription status, quantity received, and quantity remaining in stock.

The pharmacy procures its drugs from registered suppliers, and the necessary supplier information comprises supplier ID, supplier name, address, contact number, and email address. Each drug can be supplied by a maximum of two different suppliers. To facilitate the ordering process, customers must provide order-related information, including order, order description, quantity, date, and time. Customer details required for the system include customer ID, name, identification number, address (comprising street, postcode, town, and state), contact number, and email address. Furthermore, the database structure incorporates a payment entity for each customer order, featuring attributes such as payment ID, description, and payment method.

- a. Based on the above case study, represent the data requirements of the Tesla Pharmacy system as a single ER diagram. (20 MARKS)

Table 1

JENGKA PHARMACY								
Controlled Drugs Dispensary								
June Report								
Codeine Maflex								
Drug Details								
Drug ID	: XCM09801							
Drug Name	: Codeine Maflex							
Brand	: Zuelig & Mann							
Supplier ID	: M1234							
Supplier Name	: Pharma Niaga							
Cust ID	Customer Name	NRFC	Transaction ID	Term	Date	Time	Qty	Total (RM)
1358	Lina Bt. Abu	711001011358	T134	Cash	01/06/07	09.40 am	2	20.00
6247	Mohd Bin Amin	841212056247	T464	Cheque	07/06/07	08.15 pm	2	20.00
3981	Zamri Bin Ali	990606033981	T478	Cash	07/06/07	10.25 am	6	60.00
4372	Ani Binti Labu	711012014372	T333	Credit Card	05/06/07	02.35 pm	4	40.00
1358	Lina Binti Bakar	711001011358	T785	Cash	16/06/07	11.05 am	3	30.00
Total Sales								170.00

- b. By using the Index Card shown in Table 1, describe and illustrate the process of normalizing the attributes to produce a set of well-designed 3NF relations. (15 MARKS)
- c. The following outlines the process for calculating the total price of all pharmaceutical products purchased at Tesla's pharmacy:

To establish applicable discounted rates, customers are categorized into two groups: privileged members and non-members. If a privileged member purchases medicines valued at RM400 or higher, they qualify for a 10% discount off the sales price. Conversely, if the medicine's cost falls below RM400, privileged members are entitled to a 5% discount. Non-members who purchase medicines valued at RM400 or more are eligible for a 2.5% discount. In all other cases, no discounts are offered to customers.

Draw a decision tree showing the process described above. (15 MARKS)

-END OF QUESTION PAPER-